NISHANT NALAWADE

Software Engineer II • Distributed Systems & Event-Driven Architecture • LLM & Agentic AI Systems • Enterprise Scale

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years building fault-tolerant, distributed enterprise systems at Microsoft and SAP. Redesigned Invoice Capture architecture from synchronous to async event-driven pipelines - scaling capacity 3.6x and eliminating failures entirely; pattern now adopted as the org-wide standard. Applies production LLM systems, RAG pipelines, and multi-agent orchestration as engineering tools to solve hard reliability and automation problems at scale. Drives platform performance benchmarking, owns security reviews (S360/OWASP), and sets code review standards. 2025 Microsoft Global AI-ERP Hackathon winner with a productized agentic system in active customer rollout.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript (Node.js, React), Java, SQL

Cloud & Azure: Azure OpenAI, Azure Cosmos DB, Azure Data Lake, Azure Service Bus, Azure Functions, Azure Blob Storage, Azure AI Search, Dataverse, Power Automate, PySpark, Azure DevOps CI/CD

AI / ML: Azure OpenAI (GPT-4o, GPT-4), RAG (Retrieval-Augmented Generation) pipelines, Vector databases (Azure AI Search, Cosmos DB vector), Embedding models, LangChain, Semantic Kernel, Multi-agent orchestration, LLM function calling & tool use, LLM output evaluation & accuracy scoring, Agentic workflow design

Architecture: Distributed systems, Event-driven architecture, Async queue patterns, ETL pipeline design, HLD & LLD, REST APIs, Low-latency system design

Frameworks: D365 Finance & Operations, PowerApps, Dataverse

EXPERIENCE

Software Engineer II | Microsoft *Jan 2022 – Present*

Invoice Capture for Accounts Payable • High-Throughput Document Processing Platform

- **Redesigned Invoice Capture architecture using event-driven async pipelines - scaled invoice support from 330 to 1,200 lines with improved end-to-end latency and fault tolerance**, re-architecting from synchronous to async event-driven processing; pattern now adopted as the org standard.
- **Eliminated invoice processing failures (25% to 0%) and improved system reliability** by implementing fault-tolerant retry logic, LLM output quality evaluation, and fallback mechanisms - enabling enterprise-grade document complexity with zero reliability regression.
- **Built RAG-based Continuous Learning Pipeline:** Designed and built a RAG (Retrieval-Augmented Generation) based continuous learning pipeline that automatically incorporates user corrections to OCR-extracted invoice data, improving extraction accuracy over time per customer environment.
- **Resolved chronic Dataverse to F&O transfer failures (35% to less than 3%)** by engineering an auto-retry mechanism with async processing and polling, eliminating stuck-invoice incidents and increasing touchless transfers by 7%.
- **Drove platform performance benchmarking** across invoice processing throughput, API response times, and database query latency - identified and resolved key bottlenecks, presented before/after architecture comparisons to leadership, and established baseline metrics to catch future regressions.
- **Owned security reviews for the Invoice Capture** under Microsoft S360; resolved OWASP injection vulnerabilities and hardened API endpoints, ensuring compliance with enterprise security standards across all services.

- **Mentored 2 engineers**, conducted engineering interview loops, and championed GitHub Copilot adoption for code generation and review across the team: authored internal design docs and architecture diagrams presented to org leadership.
- **Built an E-Invoicing Tax Agent** - an agentic workflow using Azure OpenAI and Power Automate that autonomously monitors government tax authority websites (Brazil, Egypt, Spain), detects regulatory changes, and integrates proposed ERP configuration updates, reducing tax format errors and eliminating manual intervention.

Business Performance Analytics & Archival

- **Designed and built a modularized ETL pipeline on Azure** (Python, PySpark) to process D365 Finance data for multi-stakeholder financial analytics, enabling vendor repayment risk identification.
- **Established the foundational architecture for the Archival LTR project**, setting up the base repo and PowerApps + Dataverse groundwork that the archival team continues to build on.
- **Led cross-regional collaboration between North America and India teams** to establish data pipeline infrastructure and monitoring dashboards.

Associate Full Stack Developer | **SAP** *Nov 2017 – Jan 2022*

DSC Transportation Management

- **Built Carrier Collaboration Portal** for manufacturer-carrier contract communication; delivered across two full E2E on-premise release cycles as authorization expert.
- **Developed strategic freight procurement components** covering Demands & Rates, Collaboration Portal, and Advanced Strategic Freight Procurement; authored HLD and LLD.

Big Data Margin Assurance • S/4HANA EPPM

- **Created Alexa-SAP integration POC** using Node.js, Java, and Alexa Skills; automated QA coverage with START Tool, reducing manual testing effort significantly.
- Won Invent-2019 hackathon; contributed to Innovation Dashboard internal product.

ACHIEVEMENTS & RECOGNITION

- **2025 Winner - Microsoft Global AI-ERP Hackathon** for agentic invoice capture POC (Gen AI + OCR + master data); POC now productized and in active rollout
- **Org-wide recognition** for solving platform-scale challenges with innovative architectural approaches (async pipeline architecture and invoice capacity redesign now standard patterns for the team)
- **Peer Award 2020** - SAP DSC, for significant contribution and excellent performance
- **Star Award 2018** - SAP PSUT-ICD, for outstanding team contribution

EDUCATION

M.Tech, Software Engineering | BITS Pilani *2017 – 2021*

Software engineering lifecycle, cloud application development (AWS), testing methodologies, SAP iRPA for invoice automation.

Bachelor of Computer Applications | Maharaja Surajmal Institute *2014 – 2017*

Graduated with honours. Final project: Arduino-based Wi-Fi IR device controller.